Relative Isosingularity of Hodge Moduli

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Abstract. We give a formal deformation-theoretic argument for the étale local product structure of the relative Hodge moduli space of a smooth projective family over a complex scheme of finite type. A harmonic bundle on the central fibre yields an operator dg Lie algebra for simultaneous deformations of the variety and an integrable λ-connection. A formality diagram strictly over the Kodaira-Spencer dgla splits off the variety deformation problem and is equivariant under the full reductive stabilizer. After fixing the given relative family, the resulting Artin groupoid comparison identifies the coarse completed local ring with the invariant ring of a fixed quadratic cone tensored with the completed base. Alper's local completion theorem and relative Artin approximation give the étale statement. The argument includes singular and nonreduced bases.

Keywords: nonabelian Hodge theory; λ-connections good moduli spaces; formality.

1 Introduction

Let $f : X \to S$ be a smooth projective morphism with geometrically connected fibres, where S is a scheme of finite type over $\mathbb{C}$. Fix a relative polarization and a rank r. We consider semistable objects with vanishing rational Chern classes and write

$$\pi : \mathcal{M} = \mathcal{M}_{\mathrm{Hod}}(X/S, r) \to M = M_{\mathrm{Hod}}(X/S, r), \quad B = S \times \mathbb{A}^1. \tag{1}$$

Here $\mathcal{M}$ is the algebraic stack of relative integrable λ-connections, and M is its GIT good moduli space. We use the customary term "coarse moduli space" for M, whose closed points represent S-equivalence classes. The parameter λ is the coordinate on $\mathbb{A}^1$. No smoothness or reducedness assumption is imposed on S.

Theorem 1.1 (Relative isosingularity) Let $x \in M$ be a closed point with image $b \in B$, and let M_b denote the fibre over b. There is a pointed scheme (U, u) and a diagram over B

$$(M, x) \leftarrow (U, u) \to (M_b \times B, (x, b)), \tag{2}$$

in which both arrows are étale. Equivalently, with $R = \hat{\mathcal{O}}_{B,b}$, there is an isomorphism of complete local R-algebras

$$\hat{\mathcal{O}}_{M,x} \cong \hat{\mathcal{O}}_{B,b} \mathbin{\hat{\otimes}} \hat{\mathcal{O}}_{M_b,x}. \tag{3}$$

The completed tensor product is taken over $\mathbb{C}$, and the product on the right carries its R-algebra structure through the first factor.

For a fixed smooth projective variety, Simpson proves the corresponding étale local product statement for the Hodge moduli space over $\mathbb{A}^1$ [Sim97, §9, Theorem 9.1]; Conjecture 9.3 of the same paper formulates its relative analogue. Ono proves a product decomposition for the Kuranishi spaces of simultaneous deformations of a compact Kähler manifold and a polystable Higgs bundle with vanishing Chern classes [Ono24b]. His result concerns the Higgs parameter $\lambda = 0$ and provides the model for the comparison used here. Our argument extends the operator construction through the λ-parameter, retains the projection to the Kodaira-Spencer deformation problem, and then identifies the completed local rings of the relative good moduli spaces. The required compatibility with the given base is established at the level of Artin deformation groupoids.

The essential case is $b = (s, 0)$. Put $Y = X_s$ and choose the polystable Higgs bundle $e = (E, \theta)$ representing x. A harmonic metric on this single central fibre gives operators D' and D'' satisfying the two-types lemma. We construct a dg Lie algebra controlling simultaneous deformations of Y and its λ-connection and compare it, over the Kodaira-Spencer dgla, with a direct product whose bundle factor has zero differential. Taking the homotopy fibre over the specified formal family X/S yields a comparison preserving R. This step uses Artin coefficients and therefore applies equally to singular and nonreduced bases.

The comparison is equivariant under the full reductive stabilizer $H = \mathrm{Aut}(e)$. Restoring this stabilizer identifies the formal deformation stack with the quotient of a fixed quadratic Maurer-Cartan scheme, after extension of scalars to R. Alper's local completion comparison theorem [Alp12, Theorem 1.1] then identifies the completed local ring of M with the invariant functions on this formal quotient. This passage is necessary: the coarse moduli space does not, in general, represent isomorphism classes of object deformations over Artin rings. Relative Artin approximation converts the resulting R-algebra isomorphism into the common étale neighbourhood in Theorem 1.1. The argument retains the scheme structure of the quadratic locus throughout.

For completeness, the locus $\lambda \neq 0$ is handled by the Artin form of the relative Riemann-Hilbert correspondence. Scaling a λ-connection by λ^{-1} gives a flat connection. For every local Artin algebra A over $R_0 = \hat{\mathcal{O}}_{S,s}$, the relative analytic Poincaré lemma identifies flat connections on X_A/A with locally free A-local systems on the central topological space Y. This equivalence is natural in A and preserves full automorphism groups. Proper GAGA identifies the analytic and algebraic deformation groupoids. After adjoining the invertible λ-parameter, the formal groupoid is consequently a product over the original base, including its nilpotent directions. The good-moduli comparison of Section 6 and the approximation argument of Section 7 give the asserted étale local product there as well.

2 The central harmonic bundle and operator conventions

Choose a harmonic metric h using the nonabelian Hodge correspondence on the central fibre. On the space of all E-valued differential forms, write

$$D' = \partial_h + \theta^\dagger, \quad D'' = \overline{\partial}_E + \theta, \quad D = D' + D''. \tag{4}$$

Let ρ denote the semisimple local system associated with D. These operators satisfy

$$(D')^2 = (D'')^2 = [D', D''] = 0. \tag{5}$$

All brackets below are graded commutators with respect to total degree. On End-valued forms, D', D'' denote the induced differentials; whenever the distinction matters, the operators are first defined on E-valued forms.

Set

$$V^\bullet = A^\bullet(Y, \mathrm{End}\, E), \quad \mathfrak{t}^\bullet = A^{0,\bullet}\left(Y, T_Y^{1,0}\right). \tag{6}$$

The dgla $\mathfrak{t}$ carries the Kodaira-Spencer bracket and the differential $\overline{\partial}$. We use the sign convention in which $\overline{\partial} + [I_\varphi, \partial]$ is the deformed Dolbeault operator; reversing the sign of the Beltrami parameter gives an isomorphic model.

For $\varphi \in \mathfrak{t}^i$, let I_φ denote contraction, of degree $i - 1$, and define

$$R_\varphi = [I_\varphi, \partial_h] = [I_\varphi, D']. \tag{7}$$

The second equality follows from $I_\varphi \theta^\dagger = 0$. Let m_a denote left multiplication by an End-valued form a, and define

$$J(a, \varphi) = m_a + R_\varphi. \tag{8}$$

The map J is faithful: its first-order scalar symbol on functions recovers φ, and the remaining zeroth-order term recovers a. The action of R_φ on End-valued forms is induced by its commutators with multiplication operators.

Lemma 2.1 (Cartan identities) For $\varphi \in \mathfrak{t}^i$, we have

$$[R_\varphi, R_\psi] = R_{[\varphi,\psi]}, \quad [D', R_\varphi] = 0, \tag{9}$$

$$[D'', R_\varphi] = R_{\overline{\partial}\varphi} - m_{D'(I_\varphi \theta)}. \tag{10}$$

Proof. The first identity is the Cartan commutator formula under the condition $\partial_h^2 = 0$. The second follows from Equation 7, $(D')^2 = 0$, and the graded Jacobi identity. The third follows from

$$[D'', I_\varphi] = I_{\overline{\partial}\varphi} + (-1)^i m_{I_\varphi \theta} \tag{11}$$

together with $[D'', D'] = 0$. Equivalently, one can expand the commutators using

$$F_{\partial_h + \overline{\partial}_E} = -\big[\theta, \theta^\dagger\big], \quad \partial_h \theta = 0. \tag{12}$$

This also shows that all signs are fixed by the operator definitions, independently of the displayed bracket conventions for pairs in different references. □

3 The λ-dgla for simultaneous deformations

Put $D_\lambda = D'' + \lambda D'$, so that $D_\lambda^2 = 0$. On

$$L^i = V^i \oplus \mathfrak{t}^i \tag{13}$$

define a bracket so that J preserves graded operator commutators. Explicitly, for elements of degrees i, j, respectively, set

$$[(a, \varphi), (b, \psi)] = \big([a, b] + R_\varphi b - (-1)^{ij} R_\psi a, [\varphi, \psi]\big). \tag{14}$$

The commutator $[D_\lambda, -]$ corresponds to

$$d_\lambda(a, \varphi) = \Big((D'' + \lambda D')a - D'\big(I_\varphi \theta\big), \overline{\partial}\varphi\Big). \tag{15}$$

Proposition 3.1 (The operator model) Equation 14 and Equation 15 define a dgla over $\mathbb{C}[[\lambda]]$:

$$L_\lambda = (L[[\lambda]], d_\lambda, [-,-]). \tag{16}$$

The projection $L_\lambda \to \mathfrak{t}[[\lambda]]$ is a dgla morphism.

Proof. By Lemma 2.1, the image of J is closed under the bracket and $[D_\lambda, -]$, and

$$[D_\lambda, J(a, \varphi)] = J(d_\lambda(a, \varphi)). \tag{17}$$

The Jacobi identity, the Leibniz rule for the differential, and $d_\lambda^2 = 0$ therefore all follow from the operator algebra. The assertion about the projection is immediate from the formulas. In particular, λ adds only the term $\lambda D'$ to the differential on the End component; both the mixed differential and the mixed brackets remain unchanged. □

3.1 Geometric interpretation over arbitrary Artinian rings

Let A be a local Artinian complex algebra with maximal ideal $\mathfrak{m}_A$, and fix $\lambda \in \mathfrak{m}_A$. Take

$$\varphi \in \mathrm{MC}(\mathfrak{t} \otimes \mathfrak{m}_A). \tag{18}$$

On the fixed space of scalar-valued forms, put

$$b_\varphi = \overline{\partial} + \big[I_\varphi, \partial\big]. \tag{19}$$

Since φ has degree one, this is also $\overline{\partial} - \big[\partial, I_\varphi\big]$. The KS equation gives

$$b_\varphi^2 = 0, \quad \big[b_\varphi, \partial\big] = 0. \tag{20}$$

Lemma 3.2 (The formal Dolbeault-de Rham model) Let Y_A be the deformation of complex structure represented by φ. Then

$$\big(\mathcal{A}_Y^{p,\bullet} \otimes A, b_\varphi\big) \tag{21}$$

is a Dolbeault resolution of $\Omega^p_{Y_A/A}$, and ∂ induces the actual relative holomorphic exterior differential. Consequently,

$$\big(\mathcal{A}_Y^{\bullet,\bullet} \otimes A, b_\varphi + \lambda\partial\big) \tag{22}$$

is a resolution, for Y_A/A, of the λ-de Rham complex.

Proof. The kernel $\ker\big(b_\varphi\big)$ in bidegree $(0,0)$ is the structure sheaf of the KS deformation. Locally, induction on powers of $\mathfrak{m}_A$, using the Dolbeault lemma on the central fibre, produces formal holomorphic coordinates $w_1, \ldots, w_n$. The forms ∂w_i reduce modulo $\mathfrak{m}_A$ to the usual cotangent basis, so they still form a basis. Since $\big[b_\varphi, \partial\big] = 0$, they are also b_φ-closed. Thus the zeroth cohomology is freely generated by the forms $\partial w_{i_1} \wedge \ldots \wedge \partial w_{i_p}$ and agrees with the relative Kähler differentials. Local exactness in higher degrees follows by induction on the same finite filtration. Every step is a calculation over an Artinian ring; neither smoothness of the parameter space nor convergence of Taylor series is required. □

For $a = a^{0,1} + a^{1,0} \in V^1 \otimes \mathfrak{m}_A$, the corresponding MC operator is

$$\mathcal{D}_{\lambda,a,\varphi} = D_\lambda + m_a + R_\varphi. \tag{23}$$

Its two bidegree components are

$$\begin{aligned} B_\varphi &= \overline{\partial}_E + \lambda\theta^\dagger - \left[\partial_h, I_\varphi\right] + a^{0,1}, \\ N_\varphi &= \lambda\partial_h + \theta + a^{1,0}. \end{aligned} \tag{24}$$

It is a module operator over Equation 22. The two scalar symbols are exactly b_φ and $\lambda\partial$.

Proposition 3.3 (MC and geometric deformation groupoids) The MC equation of L_λ describes simultaneous Artinian deformations of Y and an integrable λ-connection. Degree-zero gauge describes their isomorphisms. This comparison is natural under changes of Artinian rings, up to coherent isomorphism.

Proof. Since J is faithful,

$$\mathcal{D}^2_{\lambda,a,\varphi} = 0 \Leftrightarrow d_\lambda(a,\varphi) + \frac{1}{2}[(a,\varphi),(a,\varphi)] = 0. \tag{25}$$

Decomposing by type, the left-hand side is equivalent to $B_\varphi^2 = 0$, $\left[B_\varphi, N_\varphi\right] = 0$, and $N_\varphi^2 = 0$. Lemma 3.2 and the formal Dolbeault lemma show that the first equation defines a locally free holomorphic bundle on Y_A, the second defines a holomorphic λ-connection on it, and the third expresses integrability.

Conversely, given an Artinian deformation of complex structure and a λ-connection on it, choose a formal smooth trivialization. Such trivializations of smooth bundles exist by induction along small extensions, since the corresponding sheaves of smooth sections are fine. After transporting the operator to the fixed space of forms, its scalar symbol is already determined by Y_A and λ; its difference from $D_\lambda + R_\varphi$ is an End-valued one-form. Thus the model describes every object.

The exponential of a degree-zero operator $J(u,v)$ simultaneously represents bundle gauge and a formal change of coordinates on the base manifold. Artinian coefficients make both the exponential and the BCH expansion finite, so operator conjugation agrees with Deligne gauge. Changing the auxiliary trivialization amounts precisely to changing gauge, which ensures the naturality of the groupoid comparison. □

4 Strict formality over the Kodaira-Spencer dgla

Let $K = \ker D' \subset V$ and $\mathbb{H}^\bullet = H^\bullet(V, D')$. Consider

$$\mathcal{K} = (K \oplus \mathfrak{k})[[\lambda]] \subset L_\lambda. \tag{26}$$

If $b \in K$ and $\varphi \in \mathfrak{k}^i$, then

$$R_\varphi b = (-1)^i D'\big(I_\varphi b\big). \tag{27}$$

Thus the mixed brackets preserve K, and the mixed differential also takes values in K. Since $D'|_K = 0$, the differential on $\mathcal{K}$ is independent of λ.

Define

$$q : \mathcal{K} \to (\mathbb{H}^\bullet, 0)[[\lambda]] \oplus \mathfrak{k}[[\lambda]], \quad q(a,\varphi) = ([a]_{D'}, \varphi). \tag{28}$$

The target is the direct product dgla. Equation 27 shows that q annihilates all mixed brackets; the mixed differential is D'-exact. For $a \in K$, Hodge decomposition gives $a = Pa + D'u$, so that $D''a = -D'D''u$. Hence q is also a chain map.

Proposition 4.1 (Relative formality diagram) There is a diagram of strict dgla quasi-isomorphisms

$$L_\lambda \leftarrow \mathcal{K} \xrightarrow{q} (\mathbb{H}^\bullet, 0)[[\lambda]] \oplus \mathfrak{k}[[\lambda]]. \tag{29}$$

Both arrows induce the identity on the $\mathfrak{k}[[\lambda]]$ component.

Proof. Let P be the common harmonic projector, let G be the Green operator for Δ'', and set $h = (D'')^* G$. The Kähler identities give

$$D''h + hD'' = 1 - P, \quad D'h + hD' = 0. \tag{30}$$

The quotient by the left-hand inclusion is $(V/K)[[\lambda]]$. The term $\lambda D'$ vanishes in this quotient because $\operatorname{im} D' \subset K$, so its differential is D''. Since h preserves K, it descends to the quotient; since the image of P lies in K, the induced contraction satisfies $D''h + hD'' = 1$.

The right-hand arrow is surjective on the underlying graded modules, with kernel $(\operatorname{im} D')[[\lambda]] \oplus 0$. The operator h preserves $\operatorname{im} D'$, and P vanishes on this subspace, so the kernel is likewise contractible. Both arrows are therefore quasi-isomorphisms.

Hodge decomposition splits the kernels and images used above, and Equation 30 is an explicit identity. Consequently, the required exactness persists after introducing Artin coefficients; no assertion that arbitrary nonflat base change preserves quasi-isomorphisms is used here. □

By the principle of two types, $\mathbb{H}^\bullet$ is isomorphic to $H^\bullet(Y, \operatorname{End}\rho)$ as a graded Lie algebra, with bracket given by the fixed de Rham cup-bracket. Set

$$Q = \left\{ u \in \mathbb{H}^1 : \frac{1}{2}[u,u] = 0 \text{ in } \mathbb{H}^2 \right\}. \tag{31}$$

This is the quadratic zero **scheme** in a finite-dimensional vector space, with its full scheme structure.

4.1 Equivariance under the stabilizer

Let $H = \operatorname{Aut}(E,\theta) = \operatorname{Aut}(\rho)$. This is a reductive group, with $\operatorname{Lie}(H) = \mathbb{H}^0$. Every $g \in H$ is a D-parallel End-valued zero-form, hence $\Delta_D g = 0$. The identity $\Delta_D = 2\Delta'' = 2\Delta'$ implies $D'g = D''g = 0$. Decomposing by type shows that g is compatible separately with $\partial_h, \overline{\partial}_E, \theta, \theta^\dagger$, as well as with contraction. Thus Equation 29 is H-equivariant.

Similarly, the Kähler identities express each adjoint operator as the commutator of the scalar operator Λ with the opposite differential. It follows that P, G, h also commute with H. This does not require the entire complex group H to preserve the chosen Hermitian metric.

5 Fixing the given relative family: Artin groupoids and homotopy fibres

Let $R_0 = \hat{\mathcal{O}}_{S,s}$ and $R = R_0[[\lambda]]$. For each local Artin R-algebra A, the original family determines a specified X_A/A and λ_A. Write $\operatorname{Def}_{e/B}(A)$ for the groupoid of deformations on this X_A as λ_A-connections, equipped with an identification of their special fibre with e; morphisms must preserve this marking.

Intrinsically, this groupoid is defined by

$$\operatorname{Def}_{e/B} = \operatorname{Def}_{Y,e,\lambda} \times^h_{\operatorname{Def}_Y \times \operatorname{Spf}\mathbb{C}[[\lambda]]} \operatorname{Spf} R. \tag{32}$$

The map from $\operatorname{Spf} R$ is determined by the given formal family. An object of the homotopy fibre includes an identification of its underlying deformation of the space with the original X_A. Thus automorphisms of X_A are not inadvertently quotiented out again.

Proposition 5.1 (Groupoid comparison over the original base) For every local Artin R-algebra A, there is a natural, H-equivariant equivalence of groupoids

$$\operatorname{Def}_{e/B}(A) \cong \left[\left\{ u \in \mathbb{H}^1 \otimes \mathfrak{m}_A : [u,u] = 0 \right\} / \exp\left(\mathbb{H}^0 \otimes \mathfrak{m}_A\right) \right]. \tag{33}$$

Proof. Proposition 3.3 supplies the geometric interpretation of simultaneous deformations. The comparison Equation 29 lies strictly over $\mathfrak{k}[[\lambda]]$, so it remains an equivalence after taking the homotopy fibre over the prescribed KS object. Deformation invariance under dgla quasi-isomorphisms is applied along the $\mathfrak{m}_A$-adic filtration: for each small extension, obstructions, lifts, and infinitesimal automorphisms are controlled by H^2, H^1, H^0, respectively. The comparison induces isomorphisms on these spaces, and induction yields an equivalence of groupoids. The explicit splittings also ensure the required comparison with Artin coefficients.

The dgla on the right is a direct product, so taking its fibre leaves only $\mathbb{H}^\bullet$ with zero differential. All comparisons preserve the specified R-algebra structure and are natural under changes of A; they are not merely unrelated equivalences for individual A.

Proper GAGA identifies the analytic and algebraic deformation problems at this point. Each X_A is proper; vector bundles, the principal-parts/Atiyah data defining connections, the flatness equations, and their morphisms can all be algebraized at each Artin order. Semistability of the special fibre ensures that these infinitesimal deformations lie in the semistable moduli stack under consideration. □

Generality of the base. This step does not require R_0 to be a power-series ring, nor does it extend X/S to a smooth ambient base. All harmonic analysis is performed solely on the central Kähler manifold Y. The

information carried by a singular or nonreduced base remains encoded in the specified formal family in Equation 32.

6 Formal quotients and coarse completed local rings

This section supplies an essential step in the main proof. A dgla first controls the deformation groupoid of marked objects; it does not directly represent the Artin points of the coarse moduli space.

6.1 An example distinguishing the two functors

Let $\mathbb{G}_m$ act on $\mathbb{A}^2$ with weights $(1,-1)$. Its good quotient is $\mathbb{A}^1_z$, where $z = xy$. Take $A = \mathbb{C}[\varepsilon]/(\varepsilon^2)$. Any deformation of an object whose special fibre is the origin has $x, y \in (\varepsilon)$, hence $xy = 0$. The completion of the coarse moduli space nevertheless has the Artin point $z \mapsto \varepsilon$, which does not lift to such an object deformation. Even so,

$$\mathbb{C}[[x,y]]^{\mathbb{G}_m} = \mathbb{C}[[xy]] = \hat{\mathcal{O}}_{\mathbb{A}^1,0}. \tag{34}$$

Thus one must take invariant **functions**, rather than objectwise isomorphism classes over each ring.

6.2 Restoring the full stabilizer

The gauge transformations in Equation 33 restrict to the identity on the special fibre. Forgetting the marking of the special fibre requires adjoining the action of $H(\mathbb{C})$. In characteristic zero,

$$H(A) = \exp(\mathfrak{h} \otimes \mathfrak{m}_A) \rtimes H(\mathbb{C}), \quad \mathfrak{h} = \mathrm{Lie}(H). \tag{35}$$

Every H-torsor over a local Artin complex algebra is trivial: its residual fibre has a complex point, and smoothness of H lifts that point. The full deformation groupoid is therefore the groupoid of Artin points of the corresponding quotient stack.

Set

$$T = R \mathbin{\hat{\otimes}} \hat{\mathcal{O}}_{Q,0}. \tag{36}$$

By Equation 33 and H-equivariance, we obtain an equivalence of formal completions along the residual gerbe BH:

$$\widehat{\mathcal{M}}_{BH} \cong [\mathrm{Spf}\, T/H] \cong \left[\hat{Q}_0/H\right] \times \mathrm{Spf}\, R. \tag{37}$$

Here formal completion means the full groupoid of objects over all local Artin complex algebras whose special fibre lies in BH; the equivalence is compatible with changes of rings and with the R-structure maps. This completion must be distinguished from completion along the entire fibre $\pi^{-1}(x)$.

Formal functions can be reconstructed by evaluation at the universal point over each test ring $T/\mathfrak{m}_T^n$. After forgetting the marking, functions must be invariant under $H(\mathbb{C})$. Since the H-action on every finite-order quotient is algebraic, the Zariski density of $H(\mathbb{C})$ shows that this is precisely algebraic H-invariance. The comparison above therefore concerns groupoids and natural functions; it cannot be replaced by a comparison of abstract sets of isomorphism classes.

Proposition 6.1 (The formal good-moduli comparison) Let $\mathcal{X} \to Z$ be a good moduli space morphism from a complex algebraic stack of finite type with affine diagonal. Let e be a closed point with linearly reductive stabilizer $H = \mathrm{Aut}(e)$. Let z be the image of e in Z. Suppose that all structures are over a base B, with b the image of z, and set $R = \hat{\mathcal{O}}_{B,b}$. If there is an equivalence of formal stacks preserving R,

$$\hat{\mathcal{X}}_{BH} \cong [\mathrm{Spf}\, T/H], \tag{38}$$

where T has residue field $\mathbb{C}$, carries a continuous algebraic H-action, and is a Noetherian complete local R-algebra, then

$$\hat{\mathcal{O}}_{Z,z} \cong T^H \tag{39}$$

as R-algebras.

Proof. Alper's local completion comparison theorem [Alp12, Theorem 1.1] states that, if $\mathcal{X}_n$ are the successive infinitesimal thickenings of BH, then

$$\hat{\mathcal{O}}_{Z,z} \cong \lim_n \Gamma\left(\mathcal{X}_n, \mathcal{O}_{\mathcal{X}_n}\right). \tag{40}$$

The equivalence of formal stacks identifies these formal functions. For $[\operatorname{Spf} T/H]$, the right-hand side is exactly T^H. Since the comparison preserves the morphism to $\operatorname{Spf} R$, the resulting isomorphism preserves the R-structure maps. □

For a GIT presentation, this comparison also has a concrete algebraic proof. Choose a Luna slice $[W/H]$ at the closed orbit, where w is an H-fixed point and $W /\!/ H \to Z$ is étale. Write $A_0 = \mathcal{O}(W)$, $\mathfrak{m} = \mathfrak{m}_w$, $D = A_0^H$, and $\mathfrak{n} = \mathfrak{m} \cap D$. Linear reductivity gives

$$\left(A_0/\mathfrak{m}^k\right)^H = D/\left(D \cap \mathfrak{m}^k\right). \tag{41}$$

The invariant Rees algebra

$$\bigoplus_{k\geq 0} \left(D \cap \mathfrak{m}^k\right) z^k \tag{42}$$

is finitely generated. If its positive-degree generators have degrees at most d, then

$$\mathfrak{n}^k \subset D \cap \mathfrak{m}^k \subset \mathfrak{n}^{\lceil k/d \rceil}. \tag{43}$$

The two filtrations are consequently cofinal, so that

$$\left(\widehat{A_0}_{\mathfrak{m}}\right)^H \cong \hat{D}_{\mathfrak{n}}. \tag{44}$$

Thus invariants commute with completion without any smoothness or reducedness assumption on the slice. Simpson [Sim94, Proposition 10.5] uses precisely this type of comparison.

6.3 Application to the Hodge moduli space

Applying Proposition 6.1 to Equation 37 gives

$$\hat{\mathcal{O}}_{M,x} \cong \left(R \mathbin{\hat{\otimes}} \hat{\mathcal{O}}_{Q,0}\right)^H \cong R \mathbin{\hat{\otimes}} \hat{\mathcal{O}}_{Q\,/\!/\,H,0}. \tag{45}$$

The last isomorphism uses the trivial action of H on R and the exactness of invariants for a reductive group. It can be verified on finite-order quotients and then passed to the limit using the cofinality established above. All rings retain their nilpotents.

Reducing modulo the maximal ideal of R yields the central-fibre isomorphism

$$\hat{\mathcal{O}}_{M_b,x} \cong \hat{\mathcal{O}}_{Q\,/\!/\,H,0}. \tag{46}$$

Consequently, Equation 45 is precisely Equation 3.

7 Proof of the main theorem

Set $Z = M_b \times B$ and $z = (x, b)$. Equation 3 gives

$$\hat{\mathcal{O}}_{M,x} \cong \hat{\mathcal{O}}_{Z,z} \tag{47}$$

as $\hat{\mathcal{O}}_{B,b}$-algebras. Since B is of finite type over $\mathbb{C}$, its local rings are excellent G-rings. The common étale neighbourhood consequence of relative Artin approximation [Stacks, Lemma 37.39.2] therefore provides a diagram over B

$$(M, x) \leftarrow (U, u) \rightarrow (Z, z), \tag{48}$$

with both arrows étale. This proves Theorem 1.1 at $\lambda = 0$; the case $\lambda \neq 0$ is treated in Section 1.

Noncanonicity. Neither the completed local ring isomorphism nor the common étale neighbourhood is canonical. The argument produces no global product and does not assert that the topological nonabelian Hodge correspondence is holomorphic. The product is obtained through formal deformation theory and invariant formal functions.

A Classical truncation and the cdga description

Degree zero in the full dgla encodes gauge transformations; its Chevalley-Eilenberg cdga cannot be treated as the coordinate ring of the coarse moduli space without further justification. In particular, one cannot simply write $\hat{\mathcal{O}}_{M,x} = H^0\,\mathrm{CE}(L_\lambda)$.

In the formal model obtained above, take the positive-degree sub-dgla

$$\mathfrak{n} = \mathbb{H}^{\geq 1}, \tag{49}$$

and let $C^\bullet_{\mathrm{MC}}$ denote its completed Chevalley-Eilenberg cdga, with scalars extended to R. This cdga is concentrated in nonpositive degrees: $\left(\mathbb{H}^1\right)^\vee$ supplies the degree-zero coordinates, while $\left(\mathbb{H}^2\right)^\vee$ supplies the degree-minus-one generators, whose differentials are the quadratic MC equations. Hence

$$H^0(C^\bullet_{\mathrm{MC}}) = R\left[\left[\left(\mathbb{H}^1\right)^\vee\right]\right]\Big/\left(\text{components of } \frac{1}{2}[u,u]\right) = T. \tag{50}$$

Here $R\left[\left[\left(\mathbb{H}^1\right)^\vee\right]\right]$ denotes the completed symmetric algebra on the indicated dual vector space. The correct formula for the coarse moduli ring is

$$\hat{\mathcal{O}}_{M,x} \cong \left(H^0(C^\bullet_{\mathrm{MC}})\right)^H. \tag{51}$$

The logical order is to take the classical truncation of the derived MC space first, and then take functions invariant under the full stabilizer H. This description agrees with the groupoid argument in Section 6; it does not identify Artin points of the coarse moduli space with isomorphism classes of objects.

B Further remarks

B.1 Why embedding into curve moduli is insufficient

Let $C \subset Y$ be a smooth complete-intersection curve. Even if restriction embeds the moduli space of Y into that of C, a product structure on the ambient space does not automatically make the ideal of the image constant. Restriction to a curve can also lose higher-dimensional integrability obstructions.

For example, take an abelian surface Y, the Higgs bundle $E = \mathcal{O}_Y^2$ with $\theta = 0$, and forms $\alpha, \beta \in H^0\left(Y, \Omega_Y^1\right)$ with $\alpha \wedge \beta \neq 0$. For noncommuting matrices A, B, the first-order direction

$$t(A\alpha + B\beta) \tag{52}$$

has the second-order integrability term

$$t^2[A,B]\alpha \wedge \beta. \tag{53}$$

This term vanishes after restriction to any curve. Integrability on the curve therefore does not imply integrability on the surface.

For a **fixed variety Y and a fixed embedding** $C \subset Y$, the comparison for End-valued forms is compatible with restriction. Use the restriction of the central pluriharmonic metric on C. Then D' and D'' commute with pullback. The harmonic projector need not commute with restriction, but the canonical maps from $\ker D'$ to D'-cohomology do. Thus the Hodge deformation problems on the fixed varieties have compatible formal normal forms for the restriction map,

$$Q_Y \,//\, H \to Q_C \,//\, H. \tag{54}$$

Lefschetz gives equality of the stabilizers and injectivity on H^1. The additional quadratic equations coming from $H^2(Y, \mathrm{End}\,\rho)$ must still be retained.

An embedding $C \subset Y$ varying over S requires more. There is generally no natural dgla map $\mathrm{KS}_Y \to \mathrm{KS}_C$: an ambient vector field need not be tangent to C. A relative reduction to curves would require a deformation model for the embedded pair (Y, C), compatibility of the mixed operators, and simultaneous approximation of the entire formal diagram. The main argument above does not use this reduction. In particular, it does not show that an arbitrary étale trivialization of curve moduli preserves the higher-dimensional image.

B.2 Why smoothness of the base is unnecessary

For curve families, the smooth moduli stack of curves allows an étale-local reduction to pullbacks of families over smooth bases. In higher dimensions the underlying varieties may have obstructed deformations. Embedding S in a smooth scheme does not ensure that X/S extends over it.

The present argument instead substitutes the full formal Kodaira-Spencer object of the given family into Equation 32. It does not extend the family. Consequently it needs neither a parameter-dependent harmonic metric nor an operation of taking purely holomorphic Taylor coefficients. It also does not require the holomorphicity of an isomonodromic Higgs section. All harmonic analysis takes place on the single central Kähler manifold.

B.3 The reductive-group extension

Let G be a connected complex reductive group, and work in a characteristic-class component to which the Corlette-Simpson correspondence applies. Choose a principal harmonic reduction of the central object and realize the operators in a faithful representation. Replace End E by ad P. The connection and $\theta, \theta^{\dagger}$ preserve this subbundle, so the bracket, mixed terms, and λ-differential remain closed in it. The stabilizer is the reductive centralizer $H = Z_G(\rho)$, and the argument using harmonic zero-forms again gives H-equivariance. The corresponding fixed quadratic cone is

$$Q_\rho = \left\{ u \in H^1(Y, \operatorname{ad}\rho) : \frac{1}{2}[u,u] = 0 \text{ in } H^2(Y, \operatorname{ad}\rho) \right\}. \tag{55}$$

Principal-bundle deformation theory and the good-moduli comparison provide the final steps. The detailed calculations above treat GL_r; this extension uses the usual principal nonabelian Hodge correspondence and the relevant moduli construction as additional inputs.

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